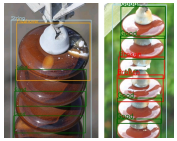
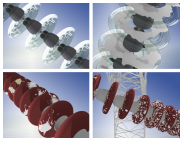
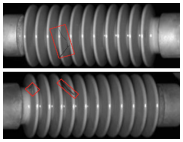
Task Description	Data				Model			
	Sample	Size	Annotations	Source	Algorithm	Architecture	Performance	Ref
Functional and Defective Discs Detection in HV Insulators Strings		1688 Images from 1618 x 1080 x 3 to 7360 x 4912 x 3	4 Classes 17,116 B. Box Num.	Open Available [47]	Supervised DL	Faster R-CNN	mAP50 = 87.2%	[221]
					Supervised DL	YOLOv8s	mAP50 = 87.9%	[221]
Defective Fraction Detection in HV Insulators Discs		1600 Images from 1618 x 1080 x 3 to 7360 x 4912 x 3	3 Classes 5,373 B. Box Num.	Open Available [222]	Supervised DL	YOLOv7	mAP50 = 90.1%	[223]
					Supervised DL	Improved YOLOv7	mAP50 = 93.8%	[223]
HV Insulators Strings and Missing Discs Detection		848 Images 1152 x 864 x 3	2 Classes 1,325 B. Box Num.	Open Available [224]	Supervised DL	YOLOv8n	mAP50 = 99.4%	[225]
					Supervised DL	SnakeNet	mAP50 = 99.5%	[225]
HV Insulators Dirty Defect Classification		47,286 Images 512 x 512 x 3 + 14,424 Images 512 x 512 x 3	4 Classes of Dirty HV Insulators	Open Available [226]	Supervised DL	Custom CNN	mIoU = 97.32%	[227]
					Supervised DL	VGG16 + RF	mAP50 = 98.54%	[228]
Catenary Insulators Defective Pixels Classification		4,293 Images 512 x 512 x 1	7 Classes 10% Defects	Data + Code Open Available [229]	"Unsupervised" DL	FastFlow [230]	F1 = 63.07%	[231]
					"Unsupervised" DL	MAE [232]	F1 = 68.89%	[231]
MV Insulators Broken Defect Detection		1,236 Images 512 x 512 x 3	1 Class 2,438 B. Box Num.	Open Available [233]	Supervised DL	YOLOv7	mAP50 = 78.5%	[234]
					Supervised DL	ID-YOLOv7	mAP50 = 85.7%	[234]
Object Detection HV Insulators		1100 Images 640 x 640 x 3	1 Class Unspecified B. Box Num.	Data + Code Open Available [235], [236]	Supervised DL	YOLOv8m	mAP50 = 92.0%	[237]
					Supervised DL	YOLOv8-Gold	mAP50 = 94.1%	[237]